

Q1 -2

Q2 -0

q3 -5b) wrong answer

Q4 -0

Q5 -0

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Differential and Integral Calculus 1M

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Homework Assignment 2

Show all your computations and justify your answers.

1. (30 points) Consider the function $f(x) = \sqrt{\frac{1}{4x^3-8}}$.(a) Show that f is one-to-one.(b) Compute $f^{-1}(x)$ and find its domain and range.(c) Compute $\lim_{x \rightarrow \frac{1}{10}} f^{-1}(x)$.

(a) Proof.

We should show that $a=b \Leftrightarrow f(a)=f(b)$ First, we should find the domain of $f(x)$ $4x^3-8 > 0$, that is $x > \sqrt[3]{2}$

$$f(a)=f(b) \Rightarrow \sqrt{\frac{1}{4a^3-8}} = \sqrt{\frac{1}{4b^3-8}}$$

$$4a^3-8 = 4b^3-8$$

$$4(a^3-b^3)=0$$

$$a^3-b^3=0$$

$$(a-b)(a^2+ab+b^2)=0$$

Because $a, b > \sqrt[3]{2}$, $a^2+ab+b^2 > 0$ then we get $a-b=0$, that is $a=b$ In conclusion, $a=b \Leftrightarrow f(a)=f(b)$

$$(b) y = \sqrt{\frac{1}{4x^3-8}}$$

$$y^2 = \frac{1}{4x^3-8}$$

$$4x^3-8 = \frac{1}{y^2}$$

$$x^3 = \frac{1+8y^2}{4y^2}$$

$$x = \sqrt[3]{\frac{1+8y^2}{4y^2}} \quad (x > \sqrt[3]{2})$$

$$\text{so } f^{-1}(x) = \sqrt[3]{\frac{1+8x^2}{4x^2}}$$

The domain is $x \in \mathbb{R} \setminus \{0\}$ The range is $f^{-1}(x) > \sqrt[3]{2}$ (c) Because in (a), we prove that f is one-to-one,

$$\begin{aligned} \text{We can write } \lim_{x \rightarrow \frac{1}{10}} f^{-1}(x) &= \sqrt[3]{\frac{1+8\left(\frac{1}{10}\right)^2}{4\left(\frac{1}{10}\right)^2}} \\ &= \sqrt[3]{\frac{1+0.08}{0.04}} \\ &= \sqrt[3]{27} \\ &= 3 \end{aligned}$$

In conclusion, $\lim_{x \rightarrow \frac{1}{10}} f^{-1}(x) = 3$

2. (21 points) Calculate the following limits.

$$(a) \lim_{x \rightarrow \sqrt{2}} \sqrt{\frac{60 - 22x^2}{2\sqrt{2} - x}}.$$

$$(b) \lim_{x \rightarrow -2} \frac{x^3 + x^2 - 2x}{x + 2}.$$

$$(c) \lim_{x \rightarrow 0} \frac{7x^2}{|x|}.$$

$$(a) \lim_{x \rightarrow \sqrt{2}} \sqrt{\frac{60 - 22x^2}{2\sqrt{2} - x}} = \sqrt{\frac{60 - 22(\sqrt{2})^2}{2\sqrt{2} - \sqrt{2}}} = \sqrt{\frac{16}{\sqrt{2}}} = \frac{4}{\sqrt{2}} = \frac{4\sqrt{2}}{2} = 2\sqrt{2}$$

$$(b) \text{ Because } -2 + 2 = 0, (-2)^3 + (-2)^2 - 2(-2) = 0$$

$$\begin{aligned} \lim_{x \rightarrow -2} \frac{x^3 + x^2 - 2x}{x + 2} &= \frac{\lim_{x \rightarrow -2} (x^3 + x^2 - 2x)'}{\lim_{x \rightarrow -2} (x + 2)'} \\ &= \frac{\lim_{x \rightarrow -2} 3x^2 + 2x - 2}{\lim_{x \rightarrow -2} 1} \\ &= \frac{3(-2)^2 + 2(-2) - 2}{1} \\ &= 6 \end{aligned}$$

$$(c) \text{ Because } 7 \cdot 0^2 = 0, |0| = 0$$

$$\begin{aligned} \text{When } x > 0, \lim_{x \rightarrow 0^+} \frac{7x^2}{|x|} &= \frac{\lim_{x \rightarrow 0^+} (7x^2)'}{\lim_{x \rightarrow 0^+} (|x|)'} = \frac{\lim_{x \rightarrow 0^+} 14x}{\lim_{x \rightarrow 0^+} (x)'} \\ &= \frac{\lim_{x \rightarrow 0^+} 14x}{\lim_{x \rightarrow 0^+} 1} \\ &= \frac{14 \times 0}{1} \\ &= 0 \end{aligned}$$

$$\text{When } x < 0, \lim_{x \rightarrow 0^-} \frac{7x^2}{|x|} = \frac{\lim_{x \rightarrow 0^-} 14x}{\lim_{x \rightarrow 0^-} (-1)} = \frac{14 \times 0}{-1} = 0$$

$$\text{According to } \lim_{x \rightarrow 0^+} \frac{7x^2}{|x|} = \lim_{x \rightarrow 0^-} \frac{7x^2}{|x|}$$

$$\text{In conclusion, } \lim_{x \rightarrow 0} \frac{7x^2}{|x|} = 0$$

3. (20 points) Given $\varepsilon = \frac{1}{5}$, find δ such that:

(a) $|x - 1| < \delta$ implies $|x^2 + x - 2| < \varepsilon$.

(b) $|x - 2| < \delta$ implies $|\frac{1}{x} - \frac{1}{2}| < \varepsilon$.

$$(a) |x^2 + x - 2| = |(x+2)(x-1)| < |(x+2)|\delta < \frac{1}{5}$$

Assume $\delta \leq 1$, then $|x+2| \leq 4$

we can write $|x+2|\delta \leq 4\delta < \frac{1}{5}$

$$\delta < \frac{1}{20}$$

In conclusion, we can know when $|x-1| < \frac{1}{20}$, $|x^2 + x - 2| < \frac{1}{5}$

$$(b) \left| \frac{1}{x} - \frac{1}{2} \right| = \left| \frac{2-x}{2x} \right| = \left| \frac{x-2}{2x} \right| = \left| \frac{1}{2x} \right| \delta < \frac{1}{5}$$

Assume $\delta \leq 1$, then $2x \leq 6$

$$\left| \frac{1}{2x} \right| \delta \leq \frac{\delta}{6} < \frac{1}{5}$$

$$\delta < \frac{6}{5}$$

In conclusion, we can know when $|x-2| < \frac{6}{5}$, $|\frac{1}{x} - \frac{1}{2}| < \frac{1}{5}$

4. (14 points) Using ε and δ definition of limits prove that $\lim_{x \rightarrow 2} \frac{1}{x} = \frac{1}{2}$

Prove: We should prove $\forall \varepsilon > 0, \exists \delta > 0, |x-2| < \delta \Rightarrow \left| \frac{1}{x} - \frac{1}{2} \right| < \varepsilon$

$$\left| \frac{1}{x} - \frac{1}{2} \right| = \left| \frac{2-x}{2x} \right| = \left| \frac{x-2}{2x} \right| = \left| \frac{1}{2x} \right| \delta < \varepsilon$$

Assume $\delta \leq 1$, then $2x \leq 6$

$$\left| \frac{1}{2x} \right| \delta \leq \frac{\delta}{6} < \varepsilon$$

$$\delta < 6\varepsilon$$



In conclusion, $\delta = \min\{1, 6\varepsilon\}$, have $|x-2| < \delta \Rightarrow \left| \frac{1}{x} - \frac{1}{2} \right| < \varepsilon$

5. (15 points) Prove that $\lim_{x \rightarrow 0} \left(\sin^2 \left(\frac{1}{x^2} \right) \sin^2(x^2) \right) = 0$.

Prove: $\lim_{x \rightarrow 0} \left(\sin^2 \left(\frac{1}{x^2} \right) \sin^2(x^2) \right)$, $\sin^2(\theta) \in [0, 1]$

When $x \rightarrow 0$, $\frac{1}{x^2} \rightarrow \infty \Rightarrow \sin^2 \left(\frac{1}{x^2} \right) \in [0, 1]$

$x \rightarrow 0$, $x^2 \rightarrow 0 \Rightarrow \sin^2(x^2) \rightarrow 0$

Then, we can write $0 \leq \sin^2 \left(\frac{1}{x^2} \right) \sin^2(x^2) \leq \sin^2(x^2)$

Since $\lim_{x \rightarrow 0} 0 = 0$, $\lim_{x \rightarrow 0} \sin^2(x^2) = \lim_{x \rightarrow 0} \sin^2(0) = 0$

Then $0 \leq \lim_{x \rightarrow 0} \left(\sin^2 \left(\frac{1}{x^2} \right) \sin^2(x^2) \right) \leq 0$

In conclusion, $\lim_{x \rightarrow 0} \left(\sin^2 \left(\frac{1}{x^2} \right) \sin^2(x^2) \right) = 0$